CALCULATION SHEETS
FOR
SUEZ CANAL TONNAGE MEASUREMENT

MALTA	BULK CARRIER	"M/V TERVEL"
NATIONALITY	KIND OF SHIP	NAME OF SHIP

OFFICIAL NUMBER	SIGNAL LETTERS	PORT OF REGISTRY	KEEL LAYING DATE
7000085	**C6R 2092**	**NASSAU**	**7 JUNE 2006**
BUILT WHERE: COCHIN, INDIA		**MATERIAL OF HULL: STEEL**	
NAME OF BUILDER: COCHIN SHIPYARD LTD		**TYPE AND NUMBER OF ENGINE: DIESEL ENGINE : ONE**	
DATE OF LAUNCH:		**KIND AND NUMBER OF PROPELLERS: SCREW PROPELLER : ONE**	

MAIN DIMENSIONS		
LENGTH	**186.100**	M
BREADTH	**32.26**	M
DEPTH	**17.80**	M

GROSS TONNAGE	**33,908.99**	**NET TONNAGE**	**32,690.12**
UNDER THE TONNAGE DECK	31,471.40	**DEDUCTED SPACES :**	
BULBOUS BOW	37.83	OFFICER'S ACCOMMODATION	260.20
FORECASTLE DECK	420.57	CREW'S ACCOMMODATION	360.83
ROUND HOUSES, ETC.	1,290.50	USED IN WORKING THE SHIP	264.29
HATCH WAYS	688.69	DEDUCTION FOR PROPELLING POWER	333.55
		TOTAL DEDUCTIONS	**1,218.87**

NOTE & CALCULATION DETAIL

SHIP RECORD: CSL HULL NO. 017 | **REGULATORY STANDARD:** *Suez Canal Authority (SCA) Rules of Navigation (Arts. 92–94 & Appendix 1)* | **UNIT:** 1 Suez Moorsom Ton = 100 cu.ft = **2.832 m³**

1. MEASUREMENT PROCEDURES EXECUTED (WHAT WAS DONE):

• **Under Deck Tonnage (UDT):** Computed via Simpson's 1st Rule over 21 transverse stations (dx = LBP / 20) with 16 vertical intervals per station (h = D / 16). Transom appendix volume aft of AP included.

• **Appendages & Enclosed Superstructures:** Bulbous bow calculated per SCA Method A (transverse cuts at St. 20, A, B and forward conical nose cap $V_{nose} = (\pi/6) \times A_B \times L_{nose}$). Forecastle deck integrated from transverse offsets. Cargo hatchways excess calculated applying 0.5% (5‰) Gross Tonnage statutory allowance. Round houses enclosed spaces schedule aggregated.

• **Gross & Net Tonnage:** Gross Tonnage obtained by summing UDT and all additions; Net Tonnage obtained by deducting statutory spaces (Officers, Crew, Navigation, and Propelling Power capped at ≤ 50% GT).

2. SUMMARY OF CALCULATED VALUES & TONNAGES (WHAT WAS OBTAINED):

• **Under Deck Tonnage (UDT):** Volume = **89,127.02 m³** ⇒ Tonnage = **31,471.40 Suez Tons**

• **Bulbous Bow:** Volume = **107.12 m³** ⇒ Tonnage = **37.83 Suez Tons** *(Fitted)*

• **Forecastle Deck:** Volume = **1,191.06 m³** ⇒ Tonnage = **420.57 Suez Tons** *(Fitted)*

• **Cargo Hatchways Excess:** Volume = **1,950.37 m³** ⇒ Tonnage = **688.69 Suez Tons** *(Excess over 0.5% GT)*

• **Round Houses / Deckhouses:** Volume = **3,654.63 m³** ⇒ Tonnage = **1,290.50 Suez Tons**

• **TOTAL SUEZ CANAL GROSS TONNAGE (GT):** Volume = **96,030.20 m³** ⇒ **Gross Tonnage = 33,908.99 Suez Tons**

• **TOTAL STATUTORY DEDUCTIONS:** Deducted Volume = **3,451.81 m³** ⇒ **Total Deductions = 1,218.87 Suez Tons**

• **FINAL SUEZ CANAL NET TONNAGE (NT):** Net Volume = **92,578.39 m³** ⇒ **Net Tonnage = 32,690.12 Suez Tons**

Underdeck Suez Tonnage Calculation — Main Hull Volume

Station	Name	Position X (m)	Depth D (m)	Area A (m²)	Factor	Product A × M
0	St. 0	0.000	17.905	101.94	1	101.94
1	St. 1	9.305	17.818	217.14	4	868.58
2	St. 2	18.610	17.800	347.66	2	695.31
3	St. 3	27.915	17.800	448.24	4	1,792.96
4	St. 4	37.220	17.800	516.48	2	1,032.96
5	St. 5	46.525	17.800	551.72	4	2,206.89
6	St. 6	55.830	17.800	565.65	2	1,131.30
7	St. 7	65.135	17.800	569.72	4	2,278.86
8	St. 8	74.440	17.800	570.65	2	1,141.31
9	St. 9	83.745	17.800	570.65	4	2,282.62
10	St. 10	93.050	17.800	570.65	2	1,141.31
11	St. 11	102.355	17.800	570.65	4	2,282.62
12	St. 12	111.660	17.800	570.65	2	1,141.31
13	St. 13	120.965	17.800	569.72	4	2,278.86
14	St. 14	130.270	17.800	565.65	2	1,131.30
15	St. 15	139.575	17.800	551.72	4	2,206.89
16	St. 16	148.880	17.800	516.48	2	1,032.96
17	St. 17	158.185	17.800	448.24	4	1,792.96
18	St. 18	167.490	17.800	402.11	2	804.22
19	St. 19	176.795	17.971	296.42	4	1,185.70
20	St. 20	186.100	18.300	74.72	1	74.72
Sum of Products: Σ = Σ (A_i × M_i)						**28,605.60**
Main Hull Volume: V_hull = (dx / 3) × Σ = (3.1017) × 28,605.60						**88,725.03 m³**
Transom Appendix Volume: V_transom (Aft of AP to St. 0)						**401.99 m³**
Total Under Deck Volume: V_UDT = V_hull + V_transom						**89,127.02 m³**
Under Deck Suez Moorsom Tonnage: UDT = V_UDT / 2.832						**31,471.40 Suez Tons**

Calculation of Transverse Station Areas

Station Transom — Depth D = 17.969 m | Interval h = D/16 = 1.1231 m | 1/3 h = 0.3744 m						
Pt	Waterline	Height Z (m)	Half-B (m)	Breadth (m)	Factor	Product B × M
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	0.000	0.000	4	0.000
3	1000WL	1.000	0.000	0.000	2	0.000
4	2000WL	2.000	0.000	0.000	4	0.000
5	3000WL	3.000	0.000	0.000	2	0.000
6	4000WL	4.000	0.000	0.000	4	0.000
7	5000WL	5.000	0.000	0.000	2	0.000
8	6000WL	6.000	0.000	0.000	4	0.000
9	7000WL	7.000	0.000	0.000	2	0.000
10	8000WL	8.000	0.000	0.000	4	0.000
11	9000WL	9.000	0.000	0.000	2	0.000
12	10000WL	10.000	0.000	0.000	4	0.000
13	11000WL	11.000	2.813	5.626	2	11.252
14	12000WL	12.000	5.425	10.850	4	43.400
15	14000WL	14.000	8.712	17.424	2	34.848
16	16000WL	16.000	11.172	22.344	4	89.376
17	Upper Deck Side	17.969	13.035	26.070	1	26.070
Sum of Products: Σ = Σ (Breadth × Factor)						**204.946**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1231 / 3) × 204.946						**76.72 m²**

Station 0 — Depth D = 17.905 m | Interval h = D/16 = 1.1191 m | 1/3 h = 0.3730 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	0.000	0.000	4	0.000
3	1000WL	1.000	0.000	0.000	2	0.000
4	2000WL	2.000	0.000	0.000	4	0.000
5	3000WL	3.000	0.000	0.000	2	0.000
6	4000WL	4.000	0.000	0.000	4	0.000
7	5000WL	5.000	0.000	0.000	2	0.000
8	6000WL	6.000	0.000	0.000	4	0.000
9	7000WL	7.000	0.000	0.000	2	0.000
10	8000WL	8.000	0.000	0.000	4	0.000
11	9000WL	9.000	0.000	0.000	2	0.000
12	10000WL	10.000	2.138	4.276	4	17.104
13	11000WL	11.000	5.735	11.470	2	22.940
14	12000WL	12.000	7.732	15.464	4	61.856
15	14000WL	14.000	10.465	20.930	2	41.860
16	16000WL	16.000	12.638	25.276	4	101.104
17	Upper Deck Side	17.905	14.211	28.422	1	28.422
Sum of Products: Σ = Σ (Breadth × Factor)						**273.286**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1191 / 3) × 273.286						**101.94 m²**

Station 0.5 — Depth D = 17.850 m | Interval h = D/16 = 1.1156 m | 1/3 h = 0.3719 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	0.000	0.000	4	0.000
3	1000WL	1.000	0.000	0.000	2	0.000
4	2000WL	2.000	0.000	0.000	4	0.000
5	3000WL	3.000	0.568	1.136	2	2.272
6	4000WL	4.000	0.000	0.000	4	0.000
7	5000WL	5.000	0.000	0.000	2	0.000
8	6000WL	6.000	0.000	0.000	4	0.000
9	7000WL	7.000	0.000	0.000	2	0.000
10	8000WL	8.000	3.989	7.978	4	31.912
11	9000WL	9.000	7.066	14.132	2	28.264
12	10000WL	10.000	8.976	17.952	4	71.808
13	11000WL	11.000	10.337	20.674	2	41.348
14	12000WL	12.000	12.424	24.848	4	99.392
15	14000WL	14.000	14.134	28.268	2	56.536
16	16000WL	16.000	15.214	30.428	4	121.712
17	Upper Deck Side	17.850	0.000	0.000	1	0.000
Sum of Products: Σ = Σ (Breadth × Factor)						**453.244**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1156 / 3) × 453.244						**168.55 m²**

Station 1 — Depth D = 17.818 m | Interval h = D/16 = 1.1136 m | 1/3 h = 0.3712 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	0.817	1.634	4	6.536
3	1000WL	1.000	1.379	2.758	2	5.516
4	2000WL	2.000	1.706	3.412	4	13.648
5	3000WL	3.000	1.638	3.276	2	6.552
6	4000WL	4.000	1.431	2.862	4	11.448
7	5000WL	5.000	1.243	2.486	2	4.972
8	6000WL	6.000	1.330	2.660	4	10.640
9	7000WL	7.000	2.766	5.532	2	11.064
10	8000WL	8.000	6.024	12.048	4	48.192
11	9000WL	9.000	8.513	17.026	2	34.052
12	10000WL	10.000	10.152	20.304	4	81.216
13	11000WL	11.000	11.359	22.718	2	45.436
14	12000WL	12.000	12.309	24.618	4	98.472
15	14000WL	14.000	13.802	27.604	2	55.208
16	16000WL	16.000	15.049	30.098	4	120.392
17	Upper Deck Side	17.818	15.810	31.620	1	31.620
Sum of Products: Σ = Σ (Breadth × Factor)						**584.964**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1136 / 3) × 584.964						**217.14 m²**

Station 1.5 — Depth D = 17.802 m | Interval h = D/16 = 1.1126 m | 1/3 h = 0.3709 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	1.936	3.872	4	15.488
3	1000WL	1.000	2.441	4.882	2	9.764
4	2000WL	2.000	2.871	5.742	4	22.968
5	3000WL	3.000	2.948	5.896	2	11.792
6	4000WL	4.000	3.031	6.062	4	24.248
7	5000WL	5.000	3.531	7.062	2	14.124
8	6000WL	6.000	5.201	10.402	4	41.608
9	7000WL	7.000	7.900	15.800	2	31.600
10	8000WL	8.000	9.662	19.324	4	77.296
11	9000WL	9.000	11.219	22.438	2	44.876
12	10000WL	10.000	12.236	24.472	4	97.888
13	11000WL	11.000	13.007	26.014	2	52.028
14	12000WL	12.000	13.634	27.268	4	109.072
15	14000WL	14.000	14.703	29.406	2	58.812
16	16000WL	16.000	15.645	31.290	4	125.160
17	Upper Deck Side	17.802	16.095	32.190	1	32.190
Sum of Products: Σ = Σ (Breadth × Factor)						**768.914**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1126 / 3) × 768.914						**285.17 m²**

Station 2 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.705	1.410	1	1.410
2	500WL	0.500	2.879	5.758	4	23.032
3	1000WL	1.000	3.536	7.072	2	14.144
4	2000WL	2.000	4.270	8.540	4	34.160
5	3000WL	3.000	4.748	9.496	2	18.992
6	4000WL	4.000	5.542	11.084	4	44.336
7	5000WL	5.000	7.402	14.804	2	29.608
8	6000WL	6.000	9.438	18.876	4	75.504
9	7000WL	7.000	11.081	22.162	2	44.324
10	8000WL	8.000	12.195	24.390	4	97.560
11	9000WL	9.000	12.978	25.956	2	51.912
12	10000WL	10.000	13.585	27.170	4	108.680
13	11000WL	11.000	14.101	28.202	2	56.404
14	12000WL	12.000	14.557	29.114	4	116.456
15	14000WL	14.000	15.324	30.648	2	61.296
16	16000WL	16.000	15.928	31.856	4	127.424
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**937.502**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 937.502						**347.66 m²**

Station 2.5 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	1.378	2.756	1	2.756
2	500WL	0.500	4.013	8.026	4	32.104
3	1000WL	1.000	4.914	9.828	2	19.656
4	2000WL	2.000	6.118	12.236	4	48.944
5	3000WL	3.000	7.362	14.724	2	29.448
6	4000WL	4.000	9.036	18.072	4	72.288
7	5000WL	5.000	10.782	21.564	2	43.128
8	6000WL	6.000	12.115	24.230	4	96.920
9	7000WL	7.000	13.014	26.028	2	52.056
10	8000WL	8.000	13.636	27.272	4	109.088
11	9000WL	9.000	14.115	28.230	2	56.460
12	10000WL	10.000	14.539	29.078	4	116.312
13	11000WL	11.000	14.912	29.824	2	59.648
14	12000WL	12.000	15.217	30.434	4	121.736
15	14000WL	14.000	15.724	31.448	2	62.896
16	16000WL	16.000	16.080	32.160	4	128.640
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1084.340**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1084.340						**402.11 m²**

Station 3 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	2.248	4.496	1	4.496
2	500WL	0.500	5.543	11.086	4	44.344
3	1000WL	1.000	6.786	13.572	2	27.144
4	2000WL	2.000	8.620	17.240	4	68.960
5	3000WL	3.000	10.329	20.658	2	41.316
6	4000WL	4.000	11.835	23.670	4	94.680
7	5000WL	5.000	12.923	25.846	2	51.692
8	6000WL	6.000	13.652	27.304	4	109.216
9	7000WL	7.000	14.177	28.354	2	56.708
10	8000WL	8.000	14.596	29.192	4	116.768
11	9000WL	9.000	14.935	29.870	2	59.740
12	10000WL	10.000	15.212	30.424	4	121.696
13	11000WL	11.000	15.437	30.874	2	61.748
14	12000WL	12.000	15.632	31.264	4	125.056
15	14000WL	14.000	15.968	31.936	2	63.872
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1208.736**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1208.736						**448.24 m²**

Station 4 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	5.064	10.128	1	10.128
2	500WL	0.500	9.965	19.930	4	79.720
3	1000WL	1.000	11.451	22.902	2	45.804
4	2000WL	2.000	13.105	26.210	4	104.840
5	3000WL	3.000	14.029	28.058	2	56.116
6	4000WL	4.000	14.605	29.210	4	116.840
7	5000WL	5.000	14.996	29.992	2	59.984
8	6000WL	6.000	15.289	30.578	4	122.312
9	7000WL	7.000	15.516	31.032	2	62.064
10	8000WL	8.000	15.689	31.378	4	125.512
11	9000WL	9.000	15.826	31.652	2	63.304
12	10000WL	10.000	15.935	31.870	4	127.480
13	11000WL	11.000	16.027	32.054	2	64.108
14	12000WL	12.000	16.091	32.182	4	128.728
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1392.760**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1392.760						**516.48 m²**

Station 5 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	9.451	18.902	1	18.902
2	500WL	0.500	13.358	26.716	4	106.864
3	1000WL	1.000	14.181	28.362	2	56.724
4	2000WL	2.000	15.037	30.074	4	120.296
5	3000WL	3.000	15.469	30.938	2	61.876
6	4000WL	4.000	15.712	31.424	4	125.696
7	5000WL	5.000	15.869	31.738	2	63.476
8	6000WL	6.000	15.983	31.966	4	127.864
9	7000WL	7.000	16.065	32.130	2	64.260
10	8000WL	8.000	16.112	32.224	4	128.896
11	9000WL	9.000	16.129	32.258	2	64.516
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1487.790**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1487.790						**551.72 m²**

Station 6 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	12.641	25.282	1	25.282
2	500WL	0.500	14.890	29.780	4	119.120
3	1000WL	1.000	15.403	30.806	2	61.612
4	2000WL	2.000	15.825	31.650	4	126.600
5	3000WL	3.000	15.991	31.982	2	63.964
6	4000WL	4.000	16.090	32.180	4	128.720
7	5000WL	5.000	16.129	32.258	2	64.516
8	6000WL	6.000	16.130	32.260	4	129.040
9	7000WL	7.000	16.130	32.260	2	64.520
10	8000WL	8.000	16.130	32.260	4	129.040
11	9000WL	9.000	16.130	32.260	2	64.520
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1525.354**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1525.354						**565.65 m²**

Station 7 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	13.888	27.776	1	27.776
2	500WL	0.500	15.362	30.724	4	122.896
3	1000WL	1.000	15.789	31.578	2	63.156
4	2000WL	2.000	16.108	32.216	4	128.864
5	3000WL	3.000	16.130	32.260	2	64.520
6	4000WL	4.000	16.130	32.260	4	129.040
7	5000WL	5.000	16.130	32.260	2	64.520
8	6000WL	6.000	16.130	32.260	4	129.040
9	7000WL	7.000	16.130	32.260	2	64.520
10	8000WL	8.000	16.130	32.260	4	129.040
11	9000WL	9.000	16.130	32.260	2	64.520
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1536.312**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1536.312						**569.72 m²**

Station 8 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	14.230	28.460	1	28.460
2	500WL	0.500	15.514	31.028	4	124.112
3	1000WL	1.000	15.903	31.806	2	63.612
4	2000WL	2.000	16.130	32.260	4	129.040
5	3000WL	3.000	16.130	32.260	2	64.520
6	4000WL	4.000	16.130	32.260	4	129.040
7	5000WL	5.000	16.130	32.260	2	64.520
8	6000WL	6.000	16.130	32.260	4	129.040
9	7000WL	7.000	16.130	32.260	2	64.520
10	8000WL	8.000	16.130	32.260	4	129.040
11	9000WL	9.000	16.130	32.260	2	64.520
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1538.844**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1538.844						**570.65 m²**

Station 9 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	14.230	28.460	1	28.460
2	500WL	0.500	15.514	31.028	4	124.112
3	1000WL	1.000	15.903	31.806	2	63.612
4	2000WL	2.000	16.130	32.260	4	129.040
5	3000WL	3.000	16.130	32.260	2	64.520
6	4000WL	4.000	16.130	32.260	4	129.040
7	5000WL	5.000	16.130	32.260	2	64.520
8	6000WL	6.000	16.130	32.260	4	129.040
9	7000WL	7.000	16.130	32.260	2	64.520
10	8000WL	8.000	16.130	32.260	4	129.040
11	9000WL	9.000	16.130	32.260	2	64.520
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1538.844**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1538.844						**570.65 m²**

Station 10 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	14.230	28.460	1	28.460
2	500WL	0.500	15.514	31.028	4	124.112
3	1000WL	1.000	15.903	31.806	2	63.612
4	2000WL	2.000	16.130	32.260	4	129.040
5	3000WL	3.000	16.130	32.260	2	64.520
6	4000WL	4.000	16.130	32.260	4	129.040
7	5000WL	5.000	16.130	32.260	2	64.520
8	6000WL	6.000	16.130	32.260	4	129.040
9	7000WL	7.000	16.130	32.260	2	64.520
10	8000WL	8.000	16.130	32.260	4	129.040
11	9000WL	9.000	16.130	32.260	2	64.520
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1538.844**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1538.844						**570.65 m²**

Station 11 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	14.230	28.460	1	28.460
2	500WL	0.500	15.514	31.028	4	124.112
3	1000WL	1.000	15.903	31.806	2	63.612
4	2000WL	2.000	16.130	32.260	4	129.040
5	3000WL	3.000	16.130	32.260	2	64.520
6	4000WL	4.000	16.130	32.260	4	129.040
7	5000WL	5.000	16.130	32.260	2	64.520
8	6000WL	6.000	16.130	32.260	4	129.040
9	7000WL	7.000	16.130	32.260	2	64.520
10	8000WL	8.000	16.130	32.260	4	129.040
11	9000WL	9.000	16.130	32.260	2	64.520
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1538.844**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1538.844						**570.65 m²**

Station 12 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	14.230	28.460	1	28.460
2	500WL	0.500	15.514	31.028	4	124.112
3	1000WL	1.000	15.903	31.806	2	63.612
4	2000WL	2.000	16.130	32.260	4	129.040
5	3000WL	3.000	16.130	32.260	2	64.520
6	4000WL	4.000	16.130	32.260	4	129.040
7	5000WL	5.000	16.130	32.260	2	64.520
8	6000WL	6.000	16.130	32.260	4	129.040
9	7000WL	7.000	16.130	32.260	2	64.520
10	8000WL	8.000	16.130	32.260	4	129.040
11	9000WL	9.000	16.130	32.260	2	64.520
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1538.844**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1538.844						**570.65 m²**

Station 13 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	13.888	27.776	1	27.776
2	500WL	0.500	15.362	30.724	4	122.896
3	1000WL	1.000	15.789	31.578	2	63.156
4	2000WL	2.000	16.108	32.216	4	128.864
5	3000WL	3.000	16.130	32.260	2	64.520
6	4000WL	4.000	16.130	32.260	4	129.040
7	5000WL	5.000	16.130	32.260	2	64.520
8	6000WL	6.000	16.130	32.260	4	129.040
9	7000WL	7.000	16.130	32.260	2	64.520
10	8000WL	8.000	16.130	32.260	4	129.040
11	9000WL	9.000	16.130	32.260	2	64.520
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1536.312**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1536.312						**569.72 m²**

Station 14 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	12.641	25.282	1	25.282
2	500WL	0.500	14.890	29.780	4	119.120
3	1000WL	1.000	15.403	30.806	2	61.612
4	2000WL	2.000	15.825	31.650	4	126.600
5	3000WL	3.000	15.991	31.982	2	63.964
6	4000WL	4.000	16.090	32.180	4	128.720
7	5000WL	5.000	16.129	32.258	2	64.516
8	6000WL	6.000	16.130	32.260	4	129.040
9	7000WL	7.000	16.130	32.260	2	64.520
10	8000WL	8.000	16.130	32.260	4	129.040
11	9000WL	9.000	16.130	32.260	2	64.520
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1525.354**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1525.354						**565.65 m²**

Station 15 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	9.451	18.902	1	18.902
2	500WL	0.500	13.358	26.716	4	106.864
3	1000WL	1.000	14.181	28.362	2	56.724
4	2000WL	2.000	15.037	30.074	4	120.296
5	3000WL	3.000	15.469	30.938	2	61.876
6	4000WL	4.000	15.712	31.424	4	125.696
7	5000WL	5.000	15.869	31.738	2	63.476
8	6000WL	6.000	15.983	31.966	4	127.864
9	7000WL	7.000	16.065	32.130	2	64.260
10	8000WL	8.000	16.112	32.224	4	128.896
11	9000WL	9.000	16.129	32.258	2	64.516
12	10000WL	10.000	16.130	32.260	4	129.040
13	11000WL	11.000	16.130	32.260	2	64.520
14	12000WL	12.000	16.130	32.260	4	129.040
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1487.790**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1487.790						**551.72 m²**

Station 16 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	5.064	10.128	1	10.128
2	500WL	0.500	9.965	19.930	4	79.720
3	1000WL	1.000	11.451	22.902	2	45.804
4	2000WL	2.000	13.105	26.210	4	104.840
5	3000WL	3.000	14.029	28.058	2	56.116
6	4000WL	4.000	14.605	29.210	4	116.840
7	5000WL	5.000	14.996	29.992	2	59.984
8	6000WL	6.000	15.289	30.578	4	122.312
9	7000WL	7.000	15.516	31.032	2	62.064
10	8000WL	8.000	15.689	31.378	4	125.512
11	9000WL	9.000	15.826	31.652	2	63.304
12	10000WL	10.000	15.935	31.870	4	127.480
13	11000WL	11.000	16.027	32.054	2	64.108
14	12000WL	12.000	16.091	32.182	4	128.728
15	14000WL	14.000	16.130	32.260	2	64.520
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1392.760**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1392.760						**516.48 m²**

Station 17 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	2.248	4.496	1	4.496
2	500WL	0.500	5.543	11.086	4	44.344
3	1000WL	1.000	6.786	13.572	2	27.144
4	2000WL	2.000	8.620	17.240	4	68.960
5	3000WL	3.000	10.329	20.658	2	41.316
6	4000WL	4.000	11.835	23.670	4	94.680
7	5000WL	5.000	12.923	25.846	2	51.692
8	6000WL	6.000	13.652	27.304	4	109.216
9	7000WL	7.000	14.177	28.354	2	56.708
10	8000WL	8.000	14.596	29.192	4	116.768
11	9000WL	9.000	14.935	29.870	2	59.740
12	10000WL	10.000	15.212	30.424	4	121.696
13	11000WL	11.000	15.437	30.874	2	61.748
14	12000WL	12.000	15.632	31.264	4	125.056
15	14000WL	14.000	15.968	31.936	2	63.872
16	16000WL	16.000	16.130	32.260	4	129.040
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1208.736**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1208.736						**448.24 m²**

Station 18 — Depth D = 17.800 m | Interval h = D/16 = 1.1125 m | 1/3 h = 0.3708 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	1.378	2.756	1	2.756
2	500WL	0.500	4.013	8.026	4	32.104
3	1000WL	1.000	4.914	9.828	2	19.656
4	2000WL	2.000	6.118	12.236	4	48.944
5	3000WL	3.000	7.362	14.724	2	29.448
6	4000WL	4.000	9.036	18.072	4	72.288
7	5000WL	5.000	10.782	21.564	2	43.128
8	6000WL	6.000	12.115	24.230	4	96.920
9	7000WL	7.000	13.014	26.028	2	52.056
10	8000WL	8.000	13.636	27.272	4	109.088
11	9000WL	9.000	14.115	28.230	2	56.460
12	10000WL	10.000	14.539	29.078	4	116.312
13	11000WL	11.000	14.912	29.824	2	59.648
14	12000WL	12.000	15.217	30.434	4	121.736
15	14000WL	14.000	15.724	31.448	2	62.896
16	16000WL	16.000	16.080	32.160	4	128.640
17	Upper Deck Side	17.800	16.130	32.260	1	32.260
Sum of Products: Σ = Σ (Breadth × Factor)						**1084.340**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1125 / 3) × 1084.340						**402.11 m²**

Station 18.5 — Depth D = 17.884 m | Interval h = D/16 = 1.1178 m | 1/3 h = 0.3726 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	3.881	7.762	4	31.048
3	1000WL	1.000	8.178	16.356	2	32.712
4	2000WL	2.000	9.517	19.034	4	76.136
5	3000WL	3.000	10.737	21.474	2	42.948
6	4000WL	4.000	11.316	22.632	4	90.528
7	5000WL	5.000	11.597	23.194	2	46.388
8	6000WL	6.000	11.684	23.368	4	93.472
9	7000WL	7.000	11.686	23.372	2	46.744
10	8000WL	8.000	11.675	23.350	4	93.400
11	9000WL	9.000	11.700	23.400	2	46.800
12	10000WL	10.000	11.776	23.552	4	94.208
13	11000WL	11.000	11.910	23.820	2	47.640
14	12000WL	12.000	12.098	24.196	4	96.784
15	14000WL	14.000	12.366	24.732	2	49.464
16	16000WL	16.000	13.050	26.100	4	104.400
17	Upper Deck Side	17.884	13.827	27.654	1	27.654
Sum of Products: Σ = Σ (Breadth × Factor)						**1020.326**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1178 / 3) × 1020.326						**380.16 m²**

Station 19 — Depth D = 17.971 m | Interval h = D/16 = 1.1232 m | 1/3 h = 0.3744 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	1.967	3.934	4	15.736
3	1000WL	1.000	5.651	11.302	2	22.604
4	2000WL	2.000	7.058	14.116	4	56.464
5	3000WL	3.000	8.372	16.744	2	33.488
6	4000WL	4.000	9.042	18.084	4	72.336
7	5000WL	5.000	9.358	18.716	2	37.432
8	6000WL	6.000	9.430	18.860	4	75.440
9	7000WL	7.000	9.341	18.682	2	37.364
10	8000WL	8.000	9.194	18.388	4	73.552
11	9000WL	9.000	9.094	18.188	2	36.376
12	10000WL	10.000	9.083	18.166	4	72.664
13	11000WL	11.000	9.167	18.334	2	36.668
14	12000WL	12.000	9.346	18.692	4	74.768
15	14000WL	14.000	9.627	19.254	2	38.508
16	16000WL	16.000	10.593	21.186	4	84.744
17	Upper Deck Side	17.971	11.798	23.596	1	23.596
Sum of Products: Σ = Σ (Breadth × Factor)						**791.740**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1232 / 3) × 791.740						**296.42 m²**

Station 19.5 — Depth D = 18.107 m | Interval h = D/16 = 1.1317 m | 1/3 h = 0.3772 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.042	0.084	1	0.084
2	500WL	0.500	2.857	5.714	4	22.856
3	1000WL	1.000	4.130	8.260	2	16.520
4	2000WL	2.000	5.533	11.066	4	44.264
5	3000WL	3.000	6.244	12.488	2	24.976
6	4000WL	4.000	6.594	13.188	4	52.752
7	5000WL	5.000	6.691	13.382	2	26.764
8	6000WL	6.000	6.569	13.138	4	52.552
9	7000WL	7.000	6.263	12.526	2	25.052
10	8000WL	8.000	5.876	11.752	4	47.008
11	9000WL	9.000	5.556	11.112	2	22.224
12	10000WL	10.000	5.404	10.808	4	43.232
13	11000WL	11.000	5.422	10.844	2	21.688
14	12000WL	12.000	5.678	11.356	4	45.424
15	14000WL	14.000	6.933	13.866	2	27.732
16	16000WL	16.000	8.645	17.290	4	69.160
17	Upper Deck Side	18.107	10.510	21.020	1	21.020
Sum of Products: Σ = Σ (Breadth × Factor)						**563.308**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1317 / 3) × 563.308						**212.50 m²**

Station 20 — Depth D = 18.300 m | Interval h = D/16 = 1.1438 m | 1/3 h = 0.3813 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	0.000	0.000	4	0.000
3	1000WL	1.000	0.889	1.778	2	3.556
4	2000WL	2.000	2.298	4.596	4	18.384
5	3000WL	3.000	3.111	6.222	2	12.444
6	4000WL	4.000	3.531	7.062	4	28.248
7	5000WL	5.000	3.659	7.318	2	14.636
8	6000WL	6.000	3.541	7.082	4	28.328
9	7000WL	7.000	3.131	6.262	2	12.524
10	8000WL	8.000	2.377	4.754	4	19.016
11	9000WL	9.000	1.282	2.564	2	5.128
12	10000WL	10.000	0.241	0.482	4	1.928
13	11000WL	11.000	0.000	0.000	2	0.000
14	12000WL	12.000	0.000	0.000	4	0.000
15	14000WL	14.000	1.666	3.332	2	6.664
16	16000WL	16.000	3.974	7.948	4	31.792
17	Upper Deck Side	18.300	6.672	13.344	1	13.344
Sum of Products: Σ = Σ (Breadth × Factor)						**195.992**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1438 / 3) × 195.992						**74.72 m²**

Station A — Depth D = 18.300 m | Interval h = D/16 = 1.1438 m | 1/3 h = 0.3813 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	0.000	0.000	4	0.000
3	1000WL	1.000	0.000	0.000	2	0.000
4	2000WL	2.000	1.172	2.344	4	9.376
5	3000WL	3.000	2.008	4.016	2	8.032
6	4000WL	4.000	2.481	4.962	4	19.848
7	5000WL	5.000	2.627	5.254	2	10.508
8	6000WL	6.000	2.517	5.034	4	20.136
9	7000WL	7.000	2.108	4.216	2	8.432
10	8000WL	8.000	1.293	2.586	4	10.344
11	9000WL	9.000	0.000	0.000	2	0.000
12	10000WL	10.000	0.000	0.000	4	0.000
13	11000WL	11.000	0.000	0.000	2	0.000
14	12000WL	12.000	0.000	0.000	4	0.000
15	14000WL	14.000	1.910	3.820	2	7.640
16	16000WL	16.000	4.819	9.638	4	38.552
17	Upper Deck Side	18.300	8.165	16.330	1	16.330
Sum of Products: Σ = Σ (Breadth × Factor)						**149.198**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1438 / 3) × 149.198						**56.88 m²**

Station B — Depth D = 18.300 m | Interval h = D/16 = 1.1438 m | 1/3 h = 0.3813 m						
Pt	**Waterline**	**Height Z (m)**	**Half-B (m)**	**Breadth (m)**	**Factor**	**Product B × M**
1	BL	0.000	0.000	0.000	1	0.000
2	500WL	0.500	0.000	0.000	4	0.000
3	1000WL	1.000	0.000	0.000	2	0.000
4	2000WL	2.000	0.870	1.740	4	6.960
5	3000WL	3.000	1.393	2.786	2	5.572
6	4000WL	4.000	1.576	3.152	4	12.608
7	5000WL	5.000	1.489	2.978	2	5.956
8	6000WL	6.000	1.088	2.176	4	8.704
9	7000WL	7.000	0.263	0.526	2	1.052
10	8000WL	8.000	0.000	0.000	4	0.000
11	9000WL	9.000	0.000	0.000	2	0.000
12	10000WL	10.000	0.000	0.000	4	0.000
13	11000WL	11.000	0.000	0.000	2	0.000
14	12000WL	12.000	0.000	0.000	4	0.000
15	14000WL	14.000	2.403	4.806	2	9.612
16	16000WL	16.000	6.287	12.574	4	50.296
17	Upper Deck Side	18.300	7.717	15.434	1	15.434
Sum of Products: Σ = Σ (Breadth × Factor)						**116.194**
Transverse Area: A_i = (h_i / 3) × Σ = (1.1438 / 3) × 116.194						**44.30 m²**

Bulbous Bow Tonnage Calculation — Method A

Station 20 Transverse Area Calculation (FP - Aft Bound of Bulb) | dy = 1.00 m						
Buttock Line	**Half-Breadth Y (m)**	**Upper Z (m)**	**Lower Z (m)**	**Depth D (m)**	**Factor**	**Product D × M**
C.L.	0.000	10.700	0.701	9.999	1.0	9.999
1000 B.L.	1.000	9.223	1.064	8.159	4.0	32.636
2000 B.L.	2.000	8.388	1.734	6.654	1.0	6.654
Sum of Products: Σ = Σ (Depth × Factor)						**49.289**
Transverse Area: A_bulb = (dy / 3) × Σ × 2 = (1.00 / 3) × 49.289 × 2						**32.859 m²**

Station A Transverse Area Calculation (Intermediate Bulb Station) | dy = 1.00 m						
Buttock Line	**Half-Breadth Y (m)**	**Upper Z (m)**	**Lower Z (m)**	**Depth D (m)**	**Factor**	**Product D × M**
C.L.	0.000	8.795	1.312	7.483	1.0	7.483
1000 B.L.	1.000	8.252	1.846	6.406	4.0	25.624
2000 B.L.	2.000	7.173	2.988	4.185	1.0	4.185
Sum of Products: Σ = Σ (Depth × Factor)						**37.292**
Transverse Area: A_bulb = (dy / 3) × Σ × 2 = (1.00 / 3) × 37.292 × 2						**24.861 m²**

Station B Transverse Area Calculation (Fore Bulb Station) | dy = 1.00 m						
Buttock Line	**Half-Breadth Y (m)**	**Upper Z (m)**	**Lower Z (m)**	**Depth D (m)**	**Factor**	**Product D × M**
C.L.	0.000	8.133	2.224	5.909	1.0	5.909
1000 B.L.	1.000	7.148	3.201	3.947	4.0	15.788
2000 B.L.	2.000	18.025	0.000	18.025	1.0	18.025
Sum of Products: Σ = Σ (Depth × Factor)						**39.722**
Transverse Area: A_bulb = (dy / 3) × Σ × 2 = (1.00 / 3) × 39.722 × 2						**26.481 m²**

Station	**Position X (m)**	**Transverse Area (m²)**	**Factor**	**Product A × M**
Station 20	0.000	32.859	1	32.859
Station A	1.500	24.861	4	99.444
Station B	3.000	26.481	1	26.481
Sum of Products: Σ = Station 20 to Station B				**158.784**
Main Bulb Body Volume: V_body = (dx / 3) × Σ = (1.50 / 3) × 158.784				**79.392 m³**
Forward Nose Cap Volume: V_nose = (π / 6) × A_B × L_nose				**27.731 m³**
Total Bulbous Bow Volume: V_bulb = V_body + V_nose				**107.123 m³**
Bulbous Bow Suez Moorsom Tonnage: T_bulb = V_bulb / 2.832				**37.83 Suez Tons**

Forecastle Tonnage Calculation

No.	**Station**	**Station Type**	**Position X (m)**	**Half-B y (m)**	**Full B (m)**	**Height H (m)**	**Area A (m²)**	**Factor**	**Product A × M**
1	Station 17	Aft Boundary	158.185	16.130	32.260	2.600	83.876	0.0	0.000
2	Station 18	Full Station	167.490	16.130	32.260	2.600	83.876	0.5	41.938
3	Station 18.5	Half Station	172.143	13.827	27.654	2.600	71.900	2.0	143.801
4	Station 19	Full Station	176.795	11.798	23.596	2.600	61.350	1.0	61.350
5	Station 19.5	Half Station	181.447	10.510	21.020	2.600	54.652	2.0	109.304
6	Station 20	FP Station	186.100	6.672	13.344	2.600	34.694	0.5	17.347
7	Station A	Stem Overhang	187.600	8.165	16.330	2.600	42.458	0.2418	10.266
8	Station B	Foremost Apex	189.100	7.717	15.434	2.600	40.128	0.0	0.000
Sum of Products: Σ_fc = Station 18 to Station A									**384.006**
Forecastle Volume: V_fc = (dx / 3) × Σ_fc = 3.1017 × 384.006									**1,191.06 m³**
Forecastle Suez Moorsom Tonnage: T_fc = V_fc / 2.832									**420.57 Suez Tons**

Cargo Hatchways Tonnage Calculation

No.	**Hatchway Description**	**Length L (m)**	**Breadth B (m)**	**Height H (m)**	**Volume V (m³)**	**Tonnage (Suez Tons)**
1	No. 1 Cargo Hatchway	15.200	14.000	1.800	383.040	135.25
2	No. 2 Cargo Hatchway	17.600	16.000	1.800	506.880	178.98
3	No. 3 Cargo Hatchway	17.600	16.000	1.800	506.880	178.98
4	No. 4 Cargo Hatchway	17.600	16.000	1.800	506.880	178.98
5	No. 5 Cargo Hatchway	17.600	16.000	1.800	506.880	178.98
Total Aggregate Hatchways: V_hatch = Σ (L × B × H)					**2,410.560 m³**	**851.19 Suez Tons**
SCA Statutory Hatch Allowance (0.5% Gross): V_allow = 0.005 × V_Gross					**460.20 m³**	**162.50 Suez Tons**
Excess Cargo Hatchways Tonnage: V_excess = max(0, V_hatch - V_allow)					**1,950.37 m³**	**688.69 Suez Tons**

Round Houses Tonnage Calculation

No.	Deck Level	Space Description	Qty	Length L (m)	Breadth B (m)	Height H (m)	Form k	Volume (m³)	Tonnage (Suez Tons)
1	Upper Platform	Steering Gear Room	1	16.61	7.09	2.50	0.880	259.083	91.48
2	Upper Platform	Purifier Room	1	9.88	9.74	2.50	0.920	221.332	78.15
3	Upper Platform	Engine Control Room	1	9.69	5.52	2.50	1.000	133.722	47.22
4	Upper Platform	Spare Parts Room	1	6.46	4.55	2.50	1.000	73.482	25.95
5	Upper Platform	Workshop	1	8.00	4.55	2.50	1.000	91.000	32.13
6	Upper Deck	CO2 Room	1	6.86	6.13	2.40	0.850	85.786	30.29
7	Upper Deck	Gymnasium / Recreation	1	6.77	3.88	2.40	1.000	63.042	22.26
8	Upper Deck	Laundry / Drying Room	1	4.81	2.25	2.40	1.000	25.974	9.17
9	Upper Deck	WC / Lavatory	1	1.96	2.25	2.40	1.000	10.584	3.74
10	Upper Deck	Deck Office	1	4.14	4.07	2.40	1.000	40.440	14.28
11	Upper Deck	Seaman Cabins	4	3.32	4.14	2.40	1.000	131.950	46.59
12	Upper Deck	Motorman Cabins	2	3.14	4.14	2.40	1.000	62.398	22.03
13	Upper Deck	A.C. Room	1	6.03	5.55	2.40	1.000	80.320	28.36
14	Upper Deck	Dry Provision Store	1	6.29	3.17	2.40	0.900	43.069	15.21
15	Upper Deck	Meat Store -18C	1	4.87	2.18	2.40	1.000	25.480	9.00
16	Upper Deck	Fish Store -18C	1	4.87	2.18	2.40	1.000	25.480	9.00
17	Upper Deck	Vegetable Store +4C	1	6.29	2.50	2.40	1.000	37.740	13.33
18	Upper Deck	Chief Engineer Office / Day Room	1	4.87	3.59	2.40	1.000	41.960	14.82
19	Boat Deck	General Store	1	1.48	6.09	2.40	1.000	21.632	7.64
20	Boat Deck	Fire-Fighting Store	1	3.14	6.09	2.40	1.000	45.894	16.21
21	Boat Deck	Boat Store	1	0.80	2.85	2.40	1.000	5.472	1.93
22	Boat Deck	Crew Mess / Lounge	1	7.89	6.06	2.40	1.000	114.752	40.52
23	Boat Deck	Hospital	1	3.94	4.30	2.40	1.000	40.661	14.36
24	Boat Deck	Motorman Cabins	2	3.95	4.30	2.40	1.000	81.528	28.79
25	Boat Deck	Galley	1	7.09	6.06	2.40	1.000	103.117	36.41
26	Boat Deck	Cook Cabin	1	3.94	4.30	2.40	1.000	40.661	14.36
27	Boat Deck	Seaman Cabins	2	3.96	4.30	2.40	1.000	81.734	28.86
28	Boat Deck	Officers' Mess / Lounge	1	4.78	6.06	2.40	1.000	69.520	24.55
29	Accommodation Deck	Seaman Cabins	3	3.33	5.26	2.40	1.000	126.114	44.53
30	Accommodation Deck	Ordinary Store	1	1.64	5.26	2.40	1.000	20.703	7.31
31	Accommodation Deck	Electrician Cabin	1	3.88	5.26	2.40	1.000	48.981	17.30
32	Accommodation Deck	Chief Motorman Cabin	1	3.94	5.26	2.40	1.000	49.739	17.56
33	Accommodation Deck	2nd Engineer Suite	1	7.86	4.33	2.40	1.000	81.681	28.84
34	Accommodation Deck	3rd Engineer Cabin	1	3.85	4.33	2.40	1.000	40.009	14.13
35	Accommodation Deck	Boatswain Cabin	1	3.91	4.33	2.40	1.000	40.633	14.35
36	Accommodation Deck	Chief Officer Suite	1	7.82	4.33	2.40	1.000	81.265	28.70
37	Captain Deck	4th Engineer Cabin	1	4.01	5.20	2.40	1.000	50.045	17.67
38	Captain Deck	Crew / Officer Office	1	2.44	5.20	2.40	1.000	30.451	10.75
39	Captain Deck	Linen / Ordinary Store	1	1.44	5.20	2.40	1.000	17.971	6.35
40	Captain Deck	Pilot Cabin	1	4.01	5.20	2.40	1.000	50.045	17.67
41	Captain Deck	3rd Officer Cabin	1	3.94	5.20	2.40	1.000	49.171	17.36
42	Captain Deck	2nd Officer Cabin	1	3.85	5.20	2.40	1.000	48.048	16.97
43	Captain Deck	Chief Engineer Suite	1	7.93	4.33	2.40	1.000	82.409	29.10
44	Captain Deck	Meeting Room	1	7.79	4.33	2.40	1.000	80.954	28.59
45	Captain Deck	Captain Suite	1	7.83	4.33	2.40	1.000	81.369	28.73
46	Bridge Deck	Radio Room	1	3.62	2.40	2.40	1.000	20.851	7.36
47	Bridge Deck	Chart Rooms (combined)	1	3.62	3.93	2.40	1.000	34.144	12.06
48	Bridge Deck	Battery Room (radio/navigation)	1	1.89	1.60	2.40	1.000	7.258	2.56
49	Bridge Deck	Ordinary Store	1	1.89	2.37	2.40	1.000	10.750	3.80
50	Bridge Deck	Wheelhouse (net)	1	15.14	5.49	2.40	1.000	199.485	70.44
51	Boat Deck	Engine Casing (measured segment)	1	7.76	7.47	2.40	1.000	139.121	49.12
52	Accom. & Capt. Decks	Funnel Casing	2	7.86	5.45	2.40	1.000	205.618	72.61
53	Unresolved	Refrigerating Machinery Room	1	0.00	0.00	0.00	1.000	0.000	0.00
Total Aggregate Round Houses: V_RH = Σ (Qty × L × B × H × k)								**3,654.630 m³**	**1,290.50 Suez Tons**
Round Houses Suez Moorsom Tonnage: T_RH = V_RH / 2.832 (Added to Gross)								**3,654.630 m³**	**1,290.50 Suez Tons**

Suez Canal Gross Tonnage Addition Calculation

Item / Component Description	Statutory Regulation	Enclosed Volume (m^3)	Suez Tonnage (Suez Tons)
Under the Tonnage Deck	SCA Rules Art. 93 Para 1	89,127.018	31,471.40
Bulbous Bow	SCA Appendix 1	107.123	37.83
Forecastle Deck	SCA Rules Art. 93 Para 2	1,191.058	420.57
Excess Cargo Hatchways	SCA Rules Art. 93 Para 3	1,950.370	688.69
Round Houses, etc.	SCA Rules Art. 93 Para 4	3,654.630	1,290.50
TOTAL SUEZ CANAL GROSS TONNAGE	**SCA Certificate Gross**	**96,030.199**	**33,908.99**

Suez Canal Net Tonnage Deduction Calculation

Statutory Deduction Space	SCA Statutory Regulation	Deducted Volume (m^3)	Deducted Tonnage (Suez Tons)
Officer's Accommodation	SCA Rules Art. 94 Para 1	736.880	260.20
Crew's Accommodation	SCA Rules Art. 94 Para 1	1,021.870	360.83
Spaces Used in Working the Ship	SCA Rules Art. 94 Para 2	748.460	264.29
Deduction for Propelling Power	SCA Rules Art. 94 Para 3	944.600	333.55
TOTAL STATUTORY DEDUCTIONS	**SCA Art. 94 Summary**	**3,451.810**	**1,218.87**
SUEZ CANAL NET TONNAGE	**SCA Certificate Net**	**-**	**32,690.12**